

# Magnetic Field Scanner (MFS) Wand

---

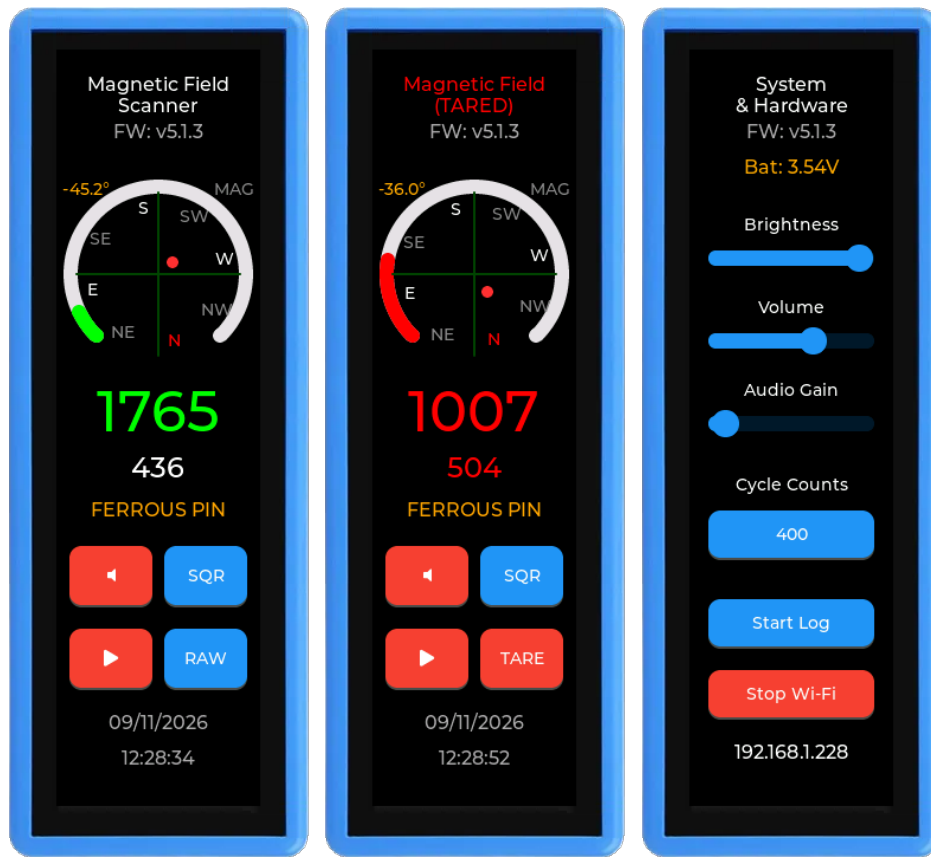
License: [GPLv3](#) Firmware: [v5.1.4](#) Platform: [ESP32-S3](#)

A professional-grade, open-source spatial magnetic gradiometer designed to detect buried ferromagnetic anomalies (property pins, pipes, UXO) by isolating highly localized magnetic gradients from the Earth's background magnetic field.

Powered by an ESP32-S3 and utilizing dual PNI RM3100 magneto-inductive sensors, this device achieves nanotesla-level sensitivity, executing 9-parameter ellipsoidal matrix calibrations and real-time Madgwick AHRS sensor fusion to provide an intuitive, HUD-style radar interface on a high-speed LCD.

---





## Hardware Configuration

### Core Components

- **Base Platform:** Waveshare ESP32-S3-Touch-LCD-3.49 v3 (PCBA v1.1 silkscreen). This highly integrated device provides the MCU, display, audio, and power management core.
- **MCU:** ESP32-S3 (16MB Flash, OPI PSRAM) embedded on the Waveshare board.
- **Magnetometers:** 2x RM3100 (TIP (at 0") and REF (at 24") sensors), with hardware reserved for a 3rd (NEAR at 8").
- **IMU:** QMI8658 (6-axis Accelerometer & Gyroscope) offset mechanically by 60° relative to the wand axis.
- **Display:** 172x640 QSPI LCD with capacitive touch.
- **Audio Codec:** ES8311 / ES7210 via I2S for audio feedback.
- **Storage:** External SPI SD Card + Internal FFat (Flash).
- **Wiring:** See the [MFS Wiring Diagram](#).
- **RTC:** PCF85063 for precise timestamping.
- **I/O Expander:** TCA9554 to offload static control pins (Backlight, Resets) and free up high-speed GPIO.

---

## Pin Allocations & Rationale

---

To support this many peripherals on a single ESP32-S3, strict pin management and an I2C I/O expander are utilized.

| Feature     | GPIO / Pin                                     | Protocol / Type | Rationale                                                                                           |
|-------------|------------------------------------------------|-----------------|-----------------------------------------------------------------------------------------------------|
| I2C Bus     | SDA: 47, SCL: 48                               | I2C             | Shared bus for RM3100s, IMU, RTC, and TCA9554 IO Expander (capped at 200kHz for cable capacitance). |
| I2S Audio   | MCLK: 7, BCLK: 15, WS: 46, DIN: 6, DOUT: 45    | I2S             | Full duplex audio codec interface.                                                                  |
| SD Card     | CS: 38, MOSI: 39, MISO: 40, SCLK: 41           | SPI             | Dedicated high-speed SPI bus for high-bandwidth data logging.                                       |
| LCD Display | CS: 9, PCLK: 10, D0-D3: 11, 12, 13, 14, TE: 21 | QSPI / 8080     | High-speed bus dedicated to driving the 172x640 LVGL display.                                       |
| Battery ADC | 4                                              | Analog (ADC1)   | Dedicated for battery voltage monitoring.                                                           |
| RM3100 DRDY | TIP: 3, REF: 2, NEAR: 5                        | Interrupt       | Direct hardware interrupts for DRDY synchronization.                                                |

| Feature    | GPIO / Pin     | Protocol / Type | Rationale                                                               |
|------------|----------------|-----------------|-------------------------------------------------------------------------|
| Future GPS | TX: 43, RX: 44 | UART            | Reserved. Freed up by migrating static control pins to the IO Expander. |

### TCA9554 I/O Expander (Address 0x20)

Used to handle low-speed/static signals to conserve MCU pins:

- **P0:** Touch Interrupt
- **P1:** LCD Backlight Enable
- **P2 / P3:** IMU Interrupts 1 & 2
- **P4:** RTC Interrupt
- **P5:** LCD Reset
- **P6:** System Power Enable (Keep-alive)
- **P7:** Night/Sleep Mode

---

## Software Architecture (v5.x.x)

The software is built on the Arduino ESP32 Core but heavily utilizes ESP-IDF native features and FreeRTOS for professional-grade isolation and performance.

### 1. Synchronized POLL Architecture & FreeRTOS

The v5.x.x architecture completely eliminates I2C collisions and phase-drift by orchestrating synchronized measurements:

- **POLL Mode:** The ESP32 broadcasts a simultaneous `REG_POLL` command to both RM3100 sensors across the I2C bus.
- **Event Groups (Core 0):** The Sensor Task sleeps at 0% CPU until both hardware `DRDY` GPIO interrupts assert, guaranteeing absolute temporal synchronization between the Tip and Reference sensors before reading.
- **LVGL UI Task (Core 1):** Handles display rendering, capacitive touch input, and real-time radar / minimap animations without interrupting the rigid I2C sensor polling pipeline.

## 2. Deep Mathematics: Kabsch & Madgwick Algorithms

To achieve military-grade spatial accuracy, the ESP32 natively executes advanced matrix mathematics and 9-axis sensor fusion:

- **Kabsch Rotational Alignment:** The system applies a 9-parameter Least-Squares Ellipsoid Fit (using a Jacobi Eigenvalue solver and the Kabsch Algorithm) to correct for Hard and Soft Iron magnetic distortions caused by the battery and LCD.
- **Madgwick 9-Axis Sensor Fusion:** The raw 3-axis Tip magnetometer data is fused with the QMI8658's 6-axis IMU (Accelerometer and Gyroscope) via an asynchronous Madgwick AHRS filter. This provides real-time Pitch, Roll, and Yaw (Azimuth) compensation to project the magnetic vector accurately to the Earth's gravity vector.
- **Universal Domain Shifting:** The `v5.1.x` architecture converts raw sensor LSBs into normalized physical nanoTeslas (nT) *before* applying the Kabsch calibration matrix. Because the matrix becomes mathematically dimensionless, a single calibration profile works universally across ALL Cycle Counts (12 to 3200).

## 3. Dynamic UI Scaling (RAW, TARE, and AUTO)

The LVGL Gradiometer UI dynamically scales its physical range and color bands based on the active mode:

- **RAW Mode (0 - 25,000 nT):** Wide dynamic range to absorb the baseline physical misalignment of the sensors (typically ~4,900 nT) without pinning the needle in the red.
- **TARE Mode (0 - 5,000 nT):** Manual zeroing of the baseline, tightening the visual arc (Green: 0-150, Yellow: 150-500, Red: >500) for extreme sensitivity to tiny localized anomalies.
- **AUTO Mode (0 - 5,000 nT):** Engages an invisible low-pass filter (triggered under 150 nT) that acts as an aggressive Auto-Tare to slowly eat away temperature drift and background anomalies while ignoring the sharp gradients of a real target.

## 4. Dual-Drive Filesystem & Integrated Web Server

- **Smart Fallback:** Dynamically routes file I/O to the high-speed SD Card, or falls back to Internal FFat if the card is missing or corrupted.
- **Captive Portal:** Hosts a Web Server allowing seamless download of `.csv` characterization logs and upload of `calibration.json` profiles.

---

## Real-World Capabilities & Detection Depth

---

The Magnetic Field Scanner is designed as a spatial gradiometer. Because it relies on an empirical RMS noise floor of  $\pm 0.33 \mu\text{T}$  (at 400 CC), it is highly sensitive to buried ferromagnetic objects.

### Use Case: Locating Property Pins (5/8" Rebar)

A standard 5/8" steel property pin driven vertically into the earth acts as a magnetic monopole, concentrating the Earth's magnetic flux at its tip.

#### Estimated Detection Depth:

- A vertical 5/8" property pin generates a surface anomaly of roughly  $500 \mu\text{T}$  at 1 inch.
- Because the wand is a gradiometer with a long handle, the near-field gradient of a monopole decays according to the inverse-cube law ( $1/r^3$ ).
- To be reliably detected above the wand's noise floor, the signal needs a minimum Signal-to-Noise Ratio (SNR).
- In a magnetically quiet environment (RAW mode), you can expect to reliably detect a standard vertical property pin buried under **1.5 to 2 feet (18 to 24 inches)** of dirt or concrete.

#### Practical Tips for Pin Locating

- **Sweep Low:** Keep the tip sensor as close to the ground as possible. Every inch of air gap costs you an inch of dirt penetration.
- **Use Audio:** Rely on the dynamic FOC audio feedback. The human ear is incredibly adept at picking up the slow frequency rise of a deep anomaly long before the UI screen indicates a massive spike.
- **Vertical vs Horizontal:** Property pins driven vertically are *much* easier to detect than pipes lying horizontally, because the vertical rod concentrates the Earth's magnetic field directly into a concentrated point (a monopole) at the surface.

---

## System Characterization

The wand features a rigid characterization methodology to empirically validate its hardware envelope across multiple Cycle Counts (CC=200, 400, 3200). To test your hardware, use the included Python script on your generated .csv logs:

```
python scripts/characterize_system.py --noise "log_noise_200.csv" "log_noise_400.csv" "log_noise_3200.csv"
```

See docs/characterization\_methodology.md for full instructions.

---

## Future Expansion Roadmap

---

The hardware architecture has been deliberately designed to accommodate future capabilities without requiring a PCB spin:

### 1. 3D Depth Sensing (3rd RM3100)

A dedicated interrupt pin (GPIO 5) and I2C address (0x22) have been reserved for a 3rd RM3100 sensor (NEAR).

- **Purpose:** Enables 2nd-order magnetic gradiometry, allowing the wand to calculate the exact depth of a magnetic anomaly (e.g., a buried pipe) rather than just its presence.

### 2. Spatial Mapping (GPS / RTK)

The static control pins for the LCD and IMU were aggressively offloaded to the TCA9554 I/O expander specifically to free up the hardware UART pins:

- **TX (GPIO 43) / RX (GPIO 44):** Reserved for a high-precision GPS or RTK module. This will allow the wand to fuse magnetic gradient data with precise geographic coordinates, generating professional magnetic survey heatmaps.

## How to Build & Deploy

---

### Prerequisites

- **Environment:** Arduino IDE v2+ or VSCode/Arduino.
- **Core:** ESP32 Board Package v3.3.x (ESP-IDF v5 based).

### Board Settings (ESP32-S3 Dev Module)

- **USB Mode:** Hardware CDC and JTAG
- **PSRAM:** OPI PSRAM
- **Flash Mode:** QIO 80MHz
- **Flash Size:** 16MB (128Mb)
- **Partition Scheme:** Custom or `app3M_fat9M_16MB`
- **Core Debug Level:** None (for performance)

### Compilation Note

All code files adhere to a strict Semantic Versioning (vx.y.z) rule enforced at every file iteration. Ensure you are building the latest `user_config.h` baseline.

## Open Source & Contributing

---

This project is open-sourced under the **GPLv3 License**. See [LICENSE](#) for details. We welcome community contributions! Please review our [Contribution Guidelines](#) before submitting pull requests.